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Inventor(s)

Ozaydin; Basak et al.

PROBABALISTIC COSTELLATION SHAPING FOR PASSIVE OPTICAL LINKS

Abstract

A method for configuring an optical transmitter for transmitting optical signals over a passive optical link. The method includes using a digital processor to generate a target constellation probability distribution (CPD) for a probabilistic constellation encoder (PCE) of the optical transmitter. The target CPD is generated using a system performance optimization algorithm accounting for the transmitter peak power constraint and a peak enhancement property of a drive circuit of the optical transmitter.

Inventors: Ozaydin; Basak (Espoo, FI), Che; Di (Espoo, FI)

Applicant: Nokia Solutions and Networks Oy (Espoo, FI)

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Background/Summary

TECHNICAL FIELD

[0001] The present invention generally relates to optical communication systems, and more

particularly, to probabilistic constellation shaping of modulation formats.

BACKGROUND

[0002] Probabilistic constellation shaping (PCS) has been deployed in commercial coherent optical transceivers, e.g., to facilitate rate adaptation to a variety of fiber channel conditions. For example, optical transceiver modules that use PCS and provide flexible data rate up to 1.2 Tb/s to cover transmission distance from tens of km to over thousands of km have been described. In coherent transmission systems over fiber-optic links with optical amplification, the launch power to the fiber may be limited by the fiber nonlinear noise, leading to an average power constraint (APC) transmission channel. It is known that for an APC channel, the optimal PCS distribution that approaches the channel capacity and maximizes the rate adaptation range is the quantized Gaussian distribution, named the Maxwell-Boltzmann (MB) distribution. The capability of rate adaptation is also attractive for short-reach interconnects without optical amplifiers. The need of such flexibility arises from the diversity of users with various rate demand and link quality. In particular, future optical access networks are expected to support a variety of user types with large disparity of channel conditions and data rate requirements.

SUMMARY

[0003] An aspect of the present disclosure provides a method that comprises generating, with a digital processor, a target constellation probability distribution (CPD) for a probabilistic constellation encoder (PCE) of an optical transmitter, and configuring the PCE of the optical transmitter based on the target CPD. The generating comprises computing the target CPD based in part on a peak power constraint on a drive signal of the optical transmitter.

[0004] A related aspect of the present disclosure provides a method that comprises generating, with a digital processor, a target CPD for a PCE of an optical transmitter, wherein the generating comprises executing a system performance optimization algorithm to search for a CPD satisfying a system performance criterion. Executing the system performance optimization algorithm comprises: determining, for a test CPD, a constellation scaling factor based on a peak power constraint of a drive signals of the optical transmitter, and computing a system performance estimate for the test CPD using the constellation scaling factor, the system performance estimate being computed for a system comprising the optical transmitter connected by a passive optical link to an optical receiver.

[0005] A related aspect of the present disclosure provides an apparatus comprising a digital processor. The digital processor is configured to generate a target CPD for a PCE of an optical transmitter. the digital processor is further configured to execute a system performance optimization algorithm to search for a CPD satisfying a system performance criterion. The system performance optimization algorithm comprises: determining, for a test CPD, a constellation scaling factor based on a peak power constraint of a drive signals of the optical transmitter; and computing a system performance estimate for the test CPD using the constellation scaling factor, the system performance estimate being computed for a system comprising the optical transmitter connected by a passive optical link to an optical receiver.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Embodiments disclosed herein will be described in greater detail with reference to the accompanying drawings which represent example embodiments thereof, which are not to scale, in which like elements are indicated with like reference numerals, and wherein:

[0007] FIG. 1 is a schematic block diagram of an optical communication system without optical amplification of PCS encoded signals according to an embodiment;

[0008] FIG. 2 is a schematic block diagram of an example probabilistic constellation encoder of an

optical transmitter of the system of FIG. 1;
[0009] FIG. 3A is a schematic diagram of a PAM symbol constellation;
[0010] FIG. 3B is a schematic diagram illustrating symbols of a PCS PAM format;
[0011] FIG. 4 is a flowchart of a method for generating a target PCS distribution for an optical transmitter of a PPC-limited optical communication system;
[0012] FIG. 5 is a flowchart of an example process for mapping a PCS distribution to a peak output of a modulator drive circuit that may be used in the method of FIG. 4;
[0013] FIG. 6 is a flowchart of an example method for generating the target PCS distribution in the method of FIG. 4 based on a mapping of a PCS distribution to a peak output of a modulator drive circuit of the optical transmitter;
[0014] FIG. 7 is a schematic functional block diagram showing elements of a system for generating a target PCS distribution using the method of FIG. 4;
[0015] FIG. 8 is a schematic block diagram of a software programmable computing system for implementing selected elements of the system of FIG. 7;
[0016] FIG. 9 is a flowchart of a method for configuring a PCE of an optical transmitter according to an example embodiment.

DETAILED DESCRIPTION

[0017] In the following description, for purposes of explanation and not limitation, specific details are set forth, such as particular circuits, circuit components, techniques, etc. in order to provide a thorough understanding of the present invention. However, it will be apparent to one skilled in the art that the present invention may be practiced in other embodiments that depart from these specific details. In other instances, detailed descriptions of well-known methods, devices, and circuits are omitted so as not to obscure the description of the present invention. All statements herein reciting principles, aspects, and embodiments of the invention, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future.

[0018] Note that as used herein, the terms “first”, “second” and so forth are not intended to imply sequential ordering, but rather are intended to distinguish one element from another, unless explicitly stated. Similarly, sequential ordering of method steps does not imply a requirement of sequential order of their execution, unless explicitly stated. The term “connected” may encompass direct connections or indirect connections through intermediate elements, unless explicitly stated otherwise.

[0019] Furthermore, the following abbreviations and acronyms may be used in the present document: [0020] APC: Average power constraint [0021] ASIC: Application specific integrated circuit [0022] BER: Bit Error Rate [0023] CPD: Constellation probability distribution [0024] D/A: Digital-to-analog [0025] DD: Direct Detection [0026] DSP: Digital signal processor [0027] E/O: Electrical to Optical [0028] FEC: Forward error correction [0029] FTTx: Fiber-to-the-X [0030] IM: Intensity modulation [0031] IM-DD: Intensity modulation and direct detection [0032] MB: Maxwell-Boltzmann [0033] O/E: Optical to Electrical [0034] QAM: Quadrature Amplitude Modulation [0035] PAM: Pulse amplitude modulation [0036] PAPR: Peak-to-average-power-ratio [0037] PAPR-E: Peak-to-average-power-ratio enhancement [0038] PCS: Probabilistic constellation shaping [0039] PMF: Probability mass function [0040] PON: Passive optical network [0041] PPC: Peak power constraint

[0042] The terms “PCS distribution” and “constellation probability distribution”, or “CPD”, are used herein interchangeably.

[0043] Examples described herein relate to optical transmitters using probabilistic constellation shaping (PCS) to communicate over passive optical links, i.e. optical links without mid-link optical amplifiers. Examples of such passive optical links include, but are not limited to, links of a passive optical network (PON) and short-reach interconnects, such as those used in datacenters among

others. The use of PCS in an optical transmitter offers a rate adaptation capability that may allow to accommodate a diversity of rate demands and/or communicate over optical links of varying quality. In particular, future optical access networks are expected to support a variety of user types with large disparity of rate demand and channel condition, such as e.g. mobile X-Haul, fiber-to-the-X (FTTx), and cable access convergence systems. Currently, transmitters of a typical PON provide a fixed rate imposed by the worst-case user in the network, leading to a waste of network capacity. [0044] The use of PCS in systems without optical amplifiers may not be as straightforward as in amplified systems with APC channels. An optical transmitter for a transmission system without optical amplifiers may be subjected to a peak power constraint (PPC), such as e.g. a limit on a peak power of a drive signal applied to an electrical-to-optical (E/O) converter (“drive signal of the (optical) transmitter”). This peak power limit, i.e. the PPC, may also be described in terms of a limit on peak amplitude of the drive signal. A PPC may arise from limitations on the extinction ratio of the E/O converter, which is typically but not exclusively an optical modulator. In an ideal PPC-limited IM transmitter having a distortion-less electrical signal channel, the PPC simply imposes an upper limit on the amplitudes of the constellation symbols, a uniform PCS distribution may be optimal. However, in real-world optical transmitters, baseband data signals go through various stages before being mapped to optical intensities by the E/O converter, e.g. as described below with reference to FIGS. 1 and 2. These stages bring about the property of “peak enhancement”, which may arise from, e.g., digital signal processing at the transmitter and bandwidth limits of various elements of a transmitter driver circuit. The peak enhancement property of optical transmitters may also be described as the enhancement of the Peak-to-Average-Power-Ratio (PAPR) of the signal from a baseband bit-to-symbol mapper to the E/O modulator, and may be referred to as the PAPR enhancement (“PAPR-E”). In a PPC limited system, the peak enhancement property of an optical transmitters complicates the selection of optimal PCS for maximizing transmitter performance. Furthermore, both the PAPR-E and the PPC may vary from transmitter to transmitter, which further complicates finding an optimum PCS.

[0045] Below, a method and an apparatus are described for selecting PCS parameters for an optical transmitter of a passive optical link that may enable maximizing link performance in a transmitter-adaptable manner.

[0046] Some aspects of configuring the PCS under PPC in the presence of peak enhancement may be described below with reference to example systems using pulse amplitude modulation (PAM), as such systems are often used in short-reach applications, especially for optical access. The skilled reader will be able to adapt the example implementations described herein to other types of PPC systems, e.g. those using coherent detection of optical PSK or QAM signals received over passive optical links, based on the description below.

[0047] FIG. 1 shows a block diagram of an optical communication system **100** (“system **100**”) having features of the present disclosure. System **100** includes an optical transmitter (“Tx”) **110** connected by an optical link **130** to an optical receiver (“Rx”) **120**. The optical link **130**, typically a fiber-optic link, is passive, i.e. lacks mid-link optical amplifiers. Accordingly, an optical signal **131** that is launched by the Tx **110** into the optical link **130** is attenuated by the propagation and coupling losses in the link, so that an optical signal **133** that is received by the Rx **120** from the link may be low in power; in such systems, the transmission performance may be limited by receiver noise, e.g. the thermal noise in the Rx **120**.

[0048] The Tx **110** includes a digital signal processor (DSP) **140**, a drive circuit **150**, and an electrical to optical (E/O) converter **115**. The DSP **140** is configured to generate a digital modulation signal **141** from an input digital signal **101** according to a selected modulation format. The E/O converter **115** is configured to generate the optical signal **131** responsive to a drive signal **151**. The drive circuit **150** is configured to convert the digital modulation signal **141** into the drive signal **151** of the E/O converter **115**.

[0049] The drive circuit **150** typically includes a digital to analog converter (DAC) **152** to convert

the digital modulation signal **141** into an analog signal, and an RF driver **154** to convert the analog output of the DAC **152** into the drive signal **151** of the E/O converter **115**.

[0050] The E/O converter **115** typically includes a laser followed by an optical modulator, which are not shown in the figure, with the drive signal **151** being applied to electrodes of the modulator. For example, the E/O converter **115** may include a distributed feedback laser integrated with an electro-absorption modulator. In another example, the optical modulator may include one or more Mach-Zehnder modulators (MZM), which may be formed in a same chip with the laser or in a different chip; e.g., a nested MZM may be used in embodiments where the Tx **110** is a coherent optical transmitter configured to use a quadrature modulation format. In other embodiments, the EO converter **115** may include a directly-modulated semiconductor laser. In any of the above implementations, there is typically a limitation, referred to herein as the PPC, on the peak power of the drive signal **151** that may be applied to the E/O converter **115**. Without loss of generality, the drive signal **151** may be referred to herein as the modulator drive signal **151**.

[0051] The Rx **120** includes an optical to electrical (O/E) converter **122**, which may be followed by an optional signal amplifier **124**, typically a TIA. The electrical output of the O/E converter **112** or the amplifier **124** is provided to a receiver DSP **128** after being digitized by an analog to digital converter (ADC) **126**. The O/E converter **122** includes one or more photodetectors, typically photodiodes, to convert the received optical signal **133** to one or more electrical signals. E.g., in embodiments configured for direct detection (DD) of PAM signals, the O/E converter **122** may include a single PD, such as a PIN diode. In embodiments with coherent optical detection, the O/E converter **122** may include an optical mixer, e.g. an 90° optical hybrid, and an optical local oscillator, as known in the art of coherent optical communications.

[0052] Referring back to the Tx **110**, the digital modulation signal **141** generated by the DSP **140** comprises a sequence of symbols of the selected modulation format, e.g. in the form of a sequence of electrical pulses of different amplitudes. A set of the symbols may be represented in a complex plane by a constellation of points defined by the modulation format, with the axes typically designated as “I” (in-phase) and “Q” (quadrature); in this representation, the distance to the center of the constellation represents the signal amplitude of the corresponding symbol. The DSP **140** is configured to use probabilistic constellation shaping (PCS) to convert the input data signal **101**, typically a binary bit stream, into the digital modulation signal **141**. The PCE **142** is configured to set the probabilities of different symbols of the selected modulation format in the modulation signal **141** according to a target probabilistic constellation shaping (PCS) distribution, which is typically described by a probability mass function (PMF). The target PCS PMF used by the PCE **142** may be adapted to the drive circuit **150** and the PPC of the E/O converter **115**, as described below. In some embodiments, the target PCS PMF used by the PCE **142** may be adapted to characteristics of the Rx **120**, e.g. the receiver noise. The Rx DSP **128** at the optical receiver **120** includes a corresponding probabilistic constellation decoder (not shown), which may operate in reverse to the PCE **142**.

[0053] FIG. 2 is a functional block diagram of an example PCE **210** that may be configured as a module of a DSP **200**. The DSP **200** may be, e.g., an ASIC, with the PCE **210** being implemented using some of the logic circuitry of the ASIC. The DSP **200** and the PCE **210** are example embodiments of the DSP **140** and the PCE **142**, respectively, of the optical transmitter **110**. The PCE **210** is configured to apply PCS to an input data signal **201** in accordance with a PCS distribution **222** to form an output digital signal **203**. The output digital signal **203** comprises a probabilistically-shaped sequence of symbols of a selected modulation format, e.g. an M-level PAM as described below. In the illustrated example, the PCE **210** includes a distribution matcher **212** and a symbol mapper **216**. The matcher **212** operates on the input data signal **210**, typically a binary bit stream, to control the probability of occurrence of different constellation symbols of the selected modulation format in the output digital signal **203** in accordance with the PCS distribution **222**. The PCS distribution **222** may be a discrete PMF described by a vector $p=[p_{\text{sub}.1}, \dots,$

p.sub.M] composed of discrete probabilities p.sub.i, i=1, . . . , M, of different constellation symbols of the selected modulation format. The matcher **212** may be configured to operate in accordance with different PCS distributions **222**, which may be provided to the matcher **212** by the DSP **200**. The symbol mapper **216** is configured to map the received bit stream to corresponding symbols of an M-symbol set, or alphabet, of the selected modulation format, with digital signals representing the symbols appearing at the output of the matcher **212** with probabilities according to the PCS distribution p **222**. An optional FEC encoder (not shown) may add forward error correction encodings to the output of the matcher **212**. The DSP **200** may include additional signal processing circuitry **218** to generate a digital input signal **205** for a modulator drive circuit. The circuitry **218** may be configured to perform some additional digital signal processing on the output **203** of the PCE **210**, such as e.g. pulse shaping, filtering, and pre-distortions to counteract inter-symbol interference (ISI). The digital modulation signal **205** is an example of the digital modulation signal **151** of the Tx **110** of FIG. **1**.

[0054] Referring back to FIG. **1**, in an ideal optical transmitter with a delta function impulse response, the peak power of the drive signal **151** would be determined by the outmost modulation level of the digital modulation signal **141** at the output of the PCE **142**. However, components of the Tx **110** are typically bandwidth-limited, and inevitably induce distortions. E.g., an eye diagram of a non-return-to-zero (NRZ) signal generated by a bandwidth-constrained transmitter typically exhibits two thicker peak levels, which is commonly referred to as peak distortion. Such distortion leads to a PCS-dependent peak power enhancement of the drive signal **151**, and to PCS-dependent PAPR-E. Furthermore, signal processing in the DSP **140** or DSP **200**, e.g. digital pulse filtering and/or pre-distortion performed by the DSP circuitry **218** downstream of the PCE **210**, may limit the signal bandwidth by digital pulse shaping, e.g. to accommodate the component bandwidth constraint. The combine effect of the digital signal distortion may be to exacerbate the PCS-dependent peak enhancement at the E/O converter input, e.g. the PAPR-E. A further description of PARP-E related effects may be found in an article by D. Che, J. Cho and X. Chen, “Does Probabilistic Constellation Shaping Benefit IM-DD Systems Without Optical Amplifiers?” Journal of Lightwave Technology, vol. 39, no. 15, pp. 4997-5007 Aug. 1, 2021, which is incorporated herein by reference.

[0055] The degree of peak enhancement or the PAPR-E depends on drive signal statistics, and thus on a particular PCS distribution, e.g. PCS PMF, **222** used by the PCE **142** or PCE **210**. Therefore, the PPC imposed on the drive signal **151** by the E/O converter **115** leads to a constraint on the baseband signal **141** that may differ for different PCS distributions **222**. This makes an optimal PCS distribution **222** for the Tx **110** dependent on PARP-E, which may differ for different Tx implementations, so that an optimal PCS distribution **222** may depend on specifics of the transmitter design. Furthermore, an optimal PCS distribution **222** may differ for different optical transmitters of nominally identical design, due to variations in characteristics of nominally identical transmitter components.

[0056] An aspect of the present disclosure provides a transmitter-dependent method for generating a PCS distribution for an optical signal transmitted over a passive optical link, such the optical signal **131** generated by the Tx **110**. The method may include i) generating a target PCS distribution for a PCE (e.g. **140** or **200**) of an optical transmitter (e.g. **110**) based in part on a transmitter PPC, and ii) configuring the PCE of the optical transmitter based on the target PCS distribution (see e.g. FIGS. **9**, at **910** and **920**). The method is described by way of example with reference to a PAM modulation format, which is typically used in short-reach IM-DD applications. The method allows a straightforward adaptation to other modulation formats, e.g. QAM, in a manner that will be evident to a skilled reader from the present description.

[0057] FIG. **3A** schematically illustrates a constellation **300** of an M-symbol PAM (“PAM constellation **300**”), with M=6 by way of example. Each symbol of the PAM alphabet is represented by black diamond aligned along the horizontal axis “I”. The distance of a symbol from

the center of the constellation represents an amplitude of an electrical signal, typically a pulse, representing the symbol in the baseband, with the position to the left or the right from zero of the horizontal axis representing the sign of the corresponding baseband electrical signal. The adjacent-level spacing Δ is illustrated at **301**, and is a “scaling” parameter of the PAM. The spacing Δ **301** may be tuned in the examples described below to scale the amplitudes of the (PAM) symbols to adapt to the PPC. The PAM **300** is bipolar symmetric in the electrical baseband; a bipolar symmetric PAM may be an optimal PAM for transmission over passive optical links, as described in the article by D. Che, J. Cho and X. Chen cited above. The drive circuit, e.g. **150**, of the E/O converter may add a DC offset to convert the bipolar baseband M-PAM **300** to an M-level intensity modulation (IM) of the drive signal **153**. In some embodiments, e.g. wherein the E/O converter includes an MZM, the drive signal **153** may be bipolar to differentially drive the MZM.

[0058] FIG. **3B** schematically illustrates a corresponding PCS PAM format **310**, resulting from applying a PCS to the PAM constellation **300**. The different heights of the bars represent PCS probabilities $p_{sub,i}$, $i=1, \dots, M$, assigned to the corresponding symbols according to an example PCS distribution, e.g. **222**. Finding an optimal, or target, PCS PAM format **310** for a particular optical transmitter or a group of transmitters may be formulated as a conditional optimization problem. Here, “optimal PCS distribution” refers to a target PCS PMF **222** that satisfies an objective criterion, e.g. maximizing, at least approximately, a transmission performance parameter of system **100**, while ensuring that the PPC imposed on the transmitter is satisfied.

[0059] Some aspects of the method of the present disclosure are illustrated below using mathematics. It will be appreciated however that the mathematical description given below is symbolic and approximate in nature, and aspects of the actual operation of corresponding devices may deviate in details from the mathematical expressions given below.

[0060] The PCS PAM format **310** used by the Tx **110** is characterized by the PCS PMF p **222** and the distance **301** Δ between adjacent PAM levels, and may be described by a tuple (Δ, p) . For the example with the symmetrical bipolar M-PAM **300**, the tuple (Δ, p) includes $M/2$ parameters, i.e. the $(M/2-1)$ independent probability values $p_{sub,i}$, and the (PAM) level spacing Δ **301**.

[0061] In the description below, “X” denotes the baseband PAM signal at the output of the PCE **142** or PCE **210**, V denotes the drive signal **151** to the E/O converter **115** (“modulator drive signal **151**”), $V_{sub,max}$ is a maximum, or peak, value of V for a given X. The signal transmission channel between the PCE **142** and the E/O converter **115** maps a peak X value, denoted $\max X$, to a peak V value, denoted $\max V$ or $V_{sub,max}$ (“peak enhancement”). The mapping, i.e. the amount or degree of the peak enhancement, depends on the PCE PMF p and the PAM level distance Δ , and may be symbolically described by the following equation (1):

$$[00001] V_{max} = f(\Delta, p, \max X) \quad (1)$$

where $f(*)$ is a function of the above indicated parameters, and may be transmitter-specific.

[0062] FIG. **4** illustrates some elements of an example method **400** for selecting a target PCS distribution p **430** for an optical transmitter of an optical transmission system subject to PPC, such as the Tx **110**. Method **400** may be executed by a computing device including digital processing hardware. In some implementations, method **400** may be executed in part or in full by digital processing hardware of the optical transmitter, such as the DSP **200** or the DSP **140** of the Tx **110**. Method **400** may also be executed in part or in full by digital processing hardware of a suitably programmed computer system that is separate from the optical transmitter; e.g., some of the related processing may be performed by a computer system of a transmitter calibration setup. The target PCS distribution p **430** selected by method **400** may then be communicated to the digital processing hardware of the optical transmitter to configure a PCE thereof, e.g. to the DSP **140** to configure the PCE **142**. The target PCS distribution p **430** may be an example of the PCE PMF **222** (FIG. **2**).

[0063] In the illustrated embodiment, method **400** begins at **410** with configuring a computing

device to map different PCS distributions p for a chosen modulation format to corresponding values of a signal peak enhancement property of the drive circuit of the optical transmitter. E.g., the computing device may be configured, e.g. programmed, to implement a p.fwdarw.V.sub.max mapping functionality **405** (“p.fwdarw.V.sub.max mapper **405**”) that maps an arbitrary PCS distribution p for the chosen modulation format, e.g. the PCS PAM **310** with some fixed $\Delta = \Delta_{\text{sub},0}$, to a corresponding peak value of the modulator drive signal, $V_{\text{sub},\text{max}}$. Method **400** at **410** accounts for the transmitter PARP-E, and the resulting mapping may be specific to a particular optical transmitter or to a group of optical transmitters having approximately same PAPR-E.

[0064] At **420**, method **400** computes a target PCS distribution **420**, denoted $p_{\text{sub},t}$, e.g. by selecting from a plurality of different PCS distributions $\{p\}$ using an optimization algorithm, based in part on a peak power constraint of the optical transmitter. The target PCS distribution **420** $p_{\text{sub},t}$ may further be configured at **420** to satisfy an objective, e.g. a transmission performance criterion.

[0065] FIG. 5 illustrates an example process **500** that may be used at **410** of method **400**, e.g. to configure the p.fwdarw.V.sub.max mapping **405**. In the illustrated embodiment, process **500** may begin at **510** with using a training set **505**, $\{p_{\text{sub},k}\}$, of PCS distributions $p_{\text{sub},k}$, $k=1, \dots, K$, to generate corresponding test inputs $X_{\text{sub},k}$ for the drive circuit of the optical transmitter, e.g. the drive circuit **150**. The number $K > 1$ of the training PCS distributions p is a parameter of the method that may be chosen differently for different transmitters, e.g. depending on the PARP-E factor of the transmitter. By way of example, for $M=8$ the K may be in a range, e.g., between 20 and 400. The training set **505** of PCS distributions, $\{p_{\text{sub},k}\}$, may be selected to sample the PCS distribution space Ω_M , e.g. in a random fashion. For the example of symmetrical bipolar M-PAM **300**, each PCS distribution $p_{\text{sub},k}$ is composed of $(M/2-1)$ independent probability values $p_{\text{sub},i}$, $i=1, \dots, (M/2-1)$. The PCS distribution space Ω_M for the M-PAM **300** includes all $p_{\text{sub},k}$ whose component probabilities $p_{\text{sub},i} \geq 0$ satisfy

$$[00002] \sum_{i=1}^{M/2} p_i = 0.5.$$

[0066] In an example implementation, generating the training set **505** of PCS distributions $p_{\text{sub},k}$ at **510** may include the following steps: i) assigning a first value in the range $[0, 0.5]$ to a probability $p_{\text{sub},1}$ for the PAM level with the smallest amplitude; ii) assigning a second value in the range $[0, 0.5 - p_{\text{sub},1}]$ to a probability $p_{\text{sub},2}$ for the PAM level with the second smallest amplitude; iii) assigning a third value in the range $[0, 0.5 - p_{\text{sub},1} - p_{\text{sub},2}]$ to a probability $p_{\text{sub},3}$ for the PAM level with the third smallest amplitude, etc. The process continues, until the level with the largest amplitude is assigned the probability

$$[00003] p_{M/2} = 0.5 - \sum_{i=1}^{M/2-1} p_i.$$

In an example implementation, the first, second, third, etc. values may be chosen randomly or pseudo-randomly in the respective ranges. This process is just one example of generating the training set of PCS distributions **505** to sample the space of PCS distributions for PAM **300**, and other implementations are possible. For example, another method to generate the training set **505** to randomly sample the space of probability distributions for M-PAM **300** may include, e.g., generating an array of random integers of size $M/2$, and normalizing the array such that the sum of the array elements is 0.5. Non-random methods of sampling the space of probability distributions for a chosen symbol constellation, e.g. with a uniform or non-uniform density of sampling points, may also be envisioned.

[0067] At **520**, process **500** may include obtaining estimates **525**, denoted $eV_{\text{sub},k}$, of peak values $V_{\text{sub},k}$ of the modulator drive signal, e.g. **153**, corresponding to the PCS distributions $p_{\text{sub},k}$ (“PCS PMFs”) of the training set **505**. This may include generating, for each of the PCS distributions $p_{\text{sub},k}$ of the training set **505**, a correspondingly shaped test sequence $X_{\text{sub},k}$ of L symbols of the chosen modulation format, e.g. PAM **300**; e.g., in each of the generated sequences, the (PAM) symbols appear with the frequencies defined by the corresponding PCS distribution $p_{\text{sub},k}$. Corresponding symbols in each of the sequences $X_{\text{sub},k}$ may be scaled to a same value,

which for the PAM **300** corresponds to having a same fixed level spacing **301**, $\Delta = \Delta_{\text{sub.0}}$. The number L of the symbols in each of the sequences $X_{\text{sub.k}}$ may be suitably large, e.g. so that each of the M symbols of the PAM alphabet appear at least once, or preferably more than once, e.g. at least 10 times, e.g. to lessen effects related to a particular choice of symbol sequences in the $X_{\text{sub.k}}$.

[0068] In some implementations, at **520** process **500** may include generating the test sequences $X_{\text{sub.k}}$ by the PCE of the optical transmitter, e.g. the PCE **142** of the Tx **110**, and measuring a peak value of a resulting modulator drive signal, e.g. **153**, at the output of the drive circuit, e.g. **150**. Corresponding peak signal measurements may be performed, e.g., at a transmitter calibration stage using an RF signal measurement tool, e.g. an oscilloscope. In some implementations, estimates $eV_{\text{sub.k}}$ of the peak drive signal may be obtained by measuring peak output optical power of the optical transmitter, e.g. using a suitably broad-band photodetector and an RF signal measurement tool.

[0069] In other implementations, a computer-implemented module may be used to obtain the peak drive signal estimates **525** $eV_{\text{sub.k}}$. The module may be configured to emulate, e.g., a frequency response of the signal channel of the optical transmitter, e.g. the Tx **110**, from the output of the PCE (e.g. **140**) to the output of the drive circuit (e.g. **150**).

[0070] To improve accuracy, subsets of several sequences $X_{\text{sub.k}}$ may be generated at **520** for each, or some, of the PCS distribution $p_{\text{sub.k}}$ of the training set **505**, and peak values of corresponding drive signals averaged over respective subsets to obtain the estimates $eV_{\text{sub.k}}$. An output of **510** of process **500** may be a training data set **527** of K tuples $(p_{\text{sub.k}}, eV_{\text{sub.k}})$.

[0071] At **530** of process **500**, the $p_{\text{fwdarw.V.sub.max}}$ mapper **405** is calibrated, or trained, using the training data set **527**. In some implementations, this may include using a curve fitting algorithm. Non-limiting examples of suitable curve-fitting algorithms include but not limited, quadratic fitting and neural network fitting. The $p_{\text{fwdarw.V.sub.max}}$ mapper **405** may be a computer-implemented function, process, algorithm, or neural network that outputs a $V_{\text{sub.max}}$ value responsive to any PCS distribution p from the space of PCS distributions of the chosen modulation format, e.g. PAM **300**.

[0072] FIG. **6** illustrates selected elements of an iterative process **600** (“process **600**”) that may be used at **420** of method **400** to obtain the target PCS distribution **430**. In the illustrated embodiment, process **600** may begin at **610** wherein constellation symbols are scaled so that a corresponding modulator drive signal, e.g. **153**, satisfies the PPC **605** of the optical transmitter, e.g. the Tx **110**, when said symbols are shaped according to a particular PCS distribution **601**. This may include providing an initial test PCS distribution **601** as input to the mapper **405** to obtain a corresponding drive signal peak estimate $eV_{\text{sub.max}}$, and scale the constellation symbols so that

$$[00004] \quad eV_{\text{max}} \leq A, \quad (2)$$

where A is a peak drive signal constraint value defined by the transmitter PPC **605**. For the PAM **300**, the scaling may comprise determining the PAM level spacing Δ **301** to satisfy the above condition (2), e.g. so that $eV_{\text{sub.max}} = A$. In an example implementation, the PAM level spacing Δ for a trial PCS PMF **601**, $\Delta(p)$, may be estimated as

$$[00005] \quad (p) = \frac{A}{eV_{\text{max}}}, \quad (3)$$

where $\Delta_{\text{sub.0}}$ is the value of the A used at **530** of process **500** to generate the mapper **405**.

[0073] For the example of PSC PAM format **310** defined by a tuple (Δ, p) , the corresponding symbol amplitudes may be expressed as

$$[00006] \quad p = \left\{ -\frac{M-1}{2} (p), \text{.Math.}, -\frac{3}{2} (p), -\frac{1}{2} (p), \frac{1}{2} (p), \frac{3}{2} (p), \text{.Math.}, \frac{M-1}{2} (p) \right\} \quad (4)$$

[0074] At **620**, the process **600** may include computing a value of an objective parameter corresponding to the PSC PAM format (Δ, p) . The objective parameter may be, e.g. a parameter related to transmission performance of a communication system subject to the PPC, e.g. system **100**. Examples of the transmission performance parameter include, but are not limited to,

transmitter-receiver mutual information in the communication system, general mutual information, bit error rate (BER) at the receiver, and the like.

[0075] In an example implementation, at **620** the process **600** may compute the mutual information $I(X;Y)$ of the communication channel between the optical transmitter, e.g. **110**, and the corresponding optical receiver, e.g. **120**. Here “X” may denote, e.g., the signal at the output of the PCE **142**, and “Y” may denote, e.g., a signal at the input of a corresponding PSC decoder of the Rx **120**. The mutual information $I(X;Y)$ quantifies the information that may be extracted at the receiver relative to the Tx-transmitted information. In an example implementation of an IM-DD channel where the receiver thermal noise is dominant, the channel can be modelled as an additive white Gaussian noise (AWGN) channel, and a conditional distribution $f_{\text{sub}.Y|X(y|x)}$ of the channel output “y” given the channel input “x” can be described by equation (5):

$$[00007] f_{Y|X}(y | x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-x)^2}{2\sigma^2}} \quad (5)$$

where a receiver noise parameter σ may be estimated, e.g., based on the signal to noise ratio (SNR) of the receiver's electrical circuitry. In a typical implementation of system **100** where the receiver noise dominates, the noise parameter σ may be estimated, e.g., based on receiver BER measurements prior to FEC. With the conditional distribution given by equation (5), the mutual information may be approximately described by the following equation (6):

[00008]

$$I(X; Y) = -\sum_{x \in \chi_{\text{sub}.p}} P(X = x) \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp(-\frac{(y-x)^2}{2\sigma^2}) \log\left(\frac{\exp(-\frac{(y-x)^2}{2\sigma^2})}{\sum_{z \in \chi_{\text{sub}.p}} P(X = z) \exp(-\frac{(y-z)^2}{2\sigma^2})}\right) dy \quad (6)$$

where the summations in the RHS of equation (6) are over all elements of the constellation vector $\chi_{\text{sub}.p}$ and $P(x)$ is the PCS probability of the corresponding element of the PCS PMF, x is an amplitude of a symbol of the $\chi_{\text{sub}.p}$ constellation.

[0076] At **620**, process **600** may estimate the mutual information for an input PCS PAM format **601**, e.g. using equation (6). In a first iteration, process **600** may proceed to **640** to update the trial PCS distribution **601**. The processing at **610** and **620** described above may then be repeated to obtain at **620** a next estimate of the objective parameter, e.g. the Tx-Rx mutual information $I(X;Y)$. At **630**, process **600** may check if a pre-defined objective criterion is satisfied. This may include, e.g., comparing a current value of the objective parameter, e.g. the mutual information, to a value or values thereof at one or more of the preceding interactions of the method, or comparing a rate of change of the objective parameter to a threshold in another example. The method may end, e.g., when no improvement is recorded, or the rate of improvement is smaller than a threshold.

[0077] The PCS distribution update at **640** may be, e.g., in accordance with one of known optimization algorithms, e.g. the gradient descent algorithm. By way of example, at **640** process **600** may estimate the gradient $\nabla_{\text{sub}.p} I(X; Y)$ of the mutual information $I(X;Y)$, e.g. using equation (6), with respect to the PCS distribution p , e.g. based on values of the corresponding probability coefficients $p_{\text{sub}.i}$ from one or more previous iterations. Process **600** may then update the PCS distribution at **640** according to, e.g., a gradient ascend update rule given by equation (7):

$$[00009] p = p - h \nabla_p I(X; Y) \quad (7)$$

The step size of the gradient ascend h may be a system parameter, and may be varied during the optimization. Starting with a large step size facilitates fast convergence, and ending up with a small step size offers better accuracy. Process **600** in this example may continue until it is determined at **630** that the gradient descent has converged, e.g. until a norm of the gradient $\nabla_{\text{sub}.p} I(X; Y)$ is determined at **630** to be smaller than a pre-defined threshold ϵ . To find a global optimum among all possible PCS PAM formats **310**, process **600** may be repeated multiple times with different initial choices of the PCS PAM **601**.

[0078] FIG. 7 illustrates selected elements of an example computing system **700** (“system **700**”) for generating a target PCS PMF for an optical transmitter of an optical communication system subject

to a PPC, e.g. the Tx **110** of system **100**. In the illustrated example, system **700** includes a test sequence generator **710**, a PCE **720**, a PMF generator **730**, a PMF to V.sub.max mapper **740**, a PPC PAM scaler **750**, a system performance estimator **760**, a Tx channel module **780**, and a memory **770** for storing modulation format parameters, e.g. a PAM PMF and a corresponding PAM level spacing Δ . In some embodiments, at least some of the shown elements of system **700** may be implemented using digital processing hardware of the optical transmitter, e.g. the DSP **140** of the Tx **120**. In an example embodiment, the PCE **720** may be an embodiment of the PCE **210** or the PCE **142** of the Tx **110**, with the memory **770** implemented within the DSP, e.g. **140** or **200**, or in a co-located memory device communicatively coupled to the DSP. The PAM PMF generator **730** may be configured to implement process **500** to generate the training set of PCS PMFs **505**, as described above with reference to FIG. 5. The PAM PMF generator **730** may also be configured to generate the trial PCS PMFs **601** in iterations of process **600**, as described above with reference to FIG. 6. The sequence generator **710** may be configured to generate the test sequences X.sub.k at **510** of process **500**, as described above with reference to FIG. 5. The Tx channel module **780** may be configured to emulate some effects of the transmission of signals generated by the PCE **720** through the electrical signal channel between the PCE and the E/O converter of the optical transmitter, e.g. between the outputs of the PCE **142** and the drive circuit **152** of the Tx **110**. E.g., the Tx channel module **780** may be configured to emulate the frequency response of the electrical signal channel of the transmitter. The Tx channel module **780** may be used by process **500** at **510** to obtain the estimates **525**, eV.sub.k, of peak values of the transmitter channel response to the training sequences X.sub.k, as described above with reference to FIG. 5. In some embodiments, wherein the PCE **720** is a part of the signal generation circuit of the optical transmitter, e.g. represents the PCE **142** of the Tx **110**, the Tx channel module **780** may be the actual signal channel of the optical transmitter for which system **700** generates the target PMF, and may include analog circuitry of the drive circuit of the E/O converter of the optical transmitter; e.g., the Tx channel module **780** may include all of the elements in the signal path from the output of the PCE **1542** to the input of the E/O converter **115**. The PPC PAM scaler **750** and the system performance estimator **760** may be configured to perform the operations at **610** and **620** of process **600**, respectively, as described above with reference to FIG. 6.

[0079] In some embodiments, at least some of the elements of system **700** shown in FIG. 7 may be implemented with hardware logic, e.g. with the digital processing hardware, of the optical transmitter. In some embodiments, at least some of the elements of system **700** shown in FIG. 7 may be implemented with software in a separate computing device, e.g. a computer system of a transmitter calibration set-up. In some embodiments, all of the elements of system **700** may be implemented in software of a computer system external to the optical transmitter. In such embodiments, parameters of the modulation format generated by the computing system **700**, e.g. the target PMF **430** and the corresponding amplitude scaling factor thereof, e.g. the PAM spacing Δ , are uploaded to the DSP of the optical transmitter, e.g. **140**, to configure the PCE thereof, e.g. the PCE **142**.

[0080] FIG. 8 illustrates selected modules of an example computing system **800** configured to generate a target PCS PMF for a PCE of an optical transmitter of a PPC-limited optical communication system, e.g. the Tx **110** of system **100**. In the illustrated example, the computing system **800** includes a memory module **810** and a processing module **820**. The memory module **810** may store processor executable instructions and/or data. In some implementations, the memory module **810** may store instructions to execute various operations of the method described herein, e.g. with reference to FIGS. 4-6. The processing module **820** is in communication with the memory module **810** and is configured to execute the instructions **805** stored in the memory module **810**. For the purposes of this disclosure, the memory module **810** may include non-transitory computer-readable media that stores data and instructions for at least a period of time. The instructions **805** may include computer code, executable by the processing module **820**, to implement all, or at least

some, of the modules **710**, **720**, **730**, **740**, **750**, **760**, and **770** of system **700** described above with reference to FIG. 7. The memory module **810** may further be configured to include at least some of: memory **812** to store the PPC of the optical transmitter, memory **814** to store data for the system performance estimator **760**, e.g. a receiver noise value, a lookup table comprising receiver noise values for different optical receivers, a lookup table comprising SNR values at the optical receiver for a plurality of link loss values, memory **816** to store Tx signal channel data for the Tx channel module **780**, e.g. the electrical frequency response of the Tx signal channel, and memory **818** for storing selected modulation format parameters, e.g. the target PMF and the corresponding A. The Tx channel data may be obtained, e.g. at a transmitter calibration stage, e.g. using an RF measurement setup, such as an oscilloscope, to measure the modulator drive signal **151** at the output of the RF drive circuit **150** (the input of the E/O converter **115**).

[0081] The memory module **810** may include, without limitation and possibly in various combinations, non-transitory storage media such as a direct access storage device (e.g., a hard disk drive or floppy disk), a sequential access storage device (e.g., a tape disk drive), compact disk (CD), random access memory (RAM), read-only memory (ROM), CD-ROM, digital versatile disc (DVD), electrically erasable programmable read-only memory (EEPROM), and flash memory.

[0082] The computing system **800** may include, or be operatively connected to, an interface module **830**. In some embodiments the interface **830** may be a suitable system, circuit, or device operable to serve as an interface between the processor **810** and digital processing hardware of an optical transmitter, such as e.g. the DSP **140**, or some units or modules thereof. The interface **830** may enable communicating a target PCS distribution, and possibly one or more related modulation format parameters to one or more optical transmitters, such as the optical transmitter **110** to configure the PCE **142**. In some implementations, the interface **830** may be communicatively coupled to a network storage resource. In some implementations, the interface **830** may be configured to interface with a person, i.e., a user. In some implementations, the interface **830** is configured to communicate with the PCE of an optical transmitter, e.g. the PCE **142** or PCE **210**, either directly or via intermediate modules or circuits of the corresponding DSP, e.g. the DSP **110** or DSP **200**.

[0083] The above described examples of the method and apparatus for configuring a PCE of an optical transmitter for communicating over a passive optical link are not intended to be limiting, and many variations will become apparent to a skilled reader having the benefit of the present disclosure. For example, the described technique may be adapted to modulation formats other than PAM; e.g., an adaptation to QAM is straightforward as QAM is composed of two PAMs mapped to the in-phase (I) and quadrature (Q) modulation signals. In some implementations, e.g. in short-interconnect optical systems with low optical loss of the transmission channel, the technique described above may account for signal noise originated at the optical transmitter (“transmitter noise”) when estimating the transmission performance of the system, e.g. at **620** of method **600**. Furthermore, the technique described above may be applicable to optical communication systems comprising one or more optical amplifiers, e.g. if transmission performance of the system is limited by the receiver noise rather than the amplified spontaneous emission (ASE) noise of optical amplifiers. Example of such systems may include systems using unsaturated optical amplifiers operating in a fixed gain mode, e.g. some booster optical amplifiers at the transmitter site. Furthermore, various features described above with reference to a specific embodiment or embodiments may be combined with other embodiments.

[0084] The examples described above, e.g. in the summary section and with reference to any one or more of the FIGS. **1-9**, provide a method (e.g. **900**, FIG. **9**). The method comprises generating, with a digital processor (e.g. **140**, FIG. **1**; **200**, FIG. **2**; **820**, FIG. **8**), a target CPD (e.g. **222**, FIG. **2**; **430**, FIGS. **4, 6, 9**) for a PCE (e.g. **142**, FIG. **1**; **210**, FIG. **2**) of an optical transmitter (e.g. **110**, FIG. **1**), and configuring the PCE of the optical transmitter based on the target CPD. The generating comprises computing the target CPD based in part on a peak power constraint (e.g. **605**, FIG. **6**) on

a drive signal (e.g. **151**, FIG. **1**) of the optical transmitter.

[0085] In some implementations, the computing may comprise selecting the target CPD from a plurality of different CPDs (e.g. **601**) based on a peak enhancement property of a drive circuit (e.g. **150**, FIG. **1**) of the optical transmitter and the peak power constraint. In some implementations, the selecting may be based on estimates (e.g. **525**, FIG. **5**) of transmission performance of an optical system (e.g. **100**, FIG. **1**) comprising a passive optical link (e.g. **130**, FIG. **1**) connecting the optical transmitter to an optical receiver (e.g. **120**, FIG. **1**) for corresponding ones of the plurality of different CPDs, and wherein the estimates are computed based on the peak enhancement property of the drive circuit of the optical transmitter and the peak power constraint. The method may comprise computing the estimates of transmission performance based in part on receiver noise of the optical receiver.

[0086] In any of the above implementations, the method may comprise mapping (e.g. at **410**, FIG. **4**) the plurality of different CPDs to values of the signal peak enhancement property (e.g. **405**). In some implementations, the mapping may include configuring the digital processor to generate an input signal (e.g. **141**, FIG. **1**) for the drive circuit for a given CPD, and to output a peak value of a corresponding output signal of the drive circuit. In some implementations, the mapping may include using a set of training CPDs (e.g. **505**, FIG. **5**) to configure the digital processor for the mapping. In some of the above implementations, the method may include providing the input signal (e.g. **141**, FIG. **1**) to a computer model (e.g. **780**, FIG. **7**) of the drive circuit of the optical transmitter. In some of the above implementations, the method may include providing the input signal to the drive circuit (e.g. **150**, FIG. **1**) of the optical transmitter.

[0087] In any of the above implementations of the method may include executing a conditional search algorithm (e.g. **600**, FIG. **6**) to generate at least some of the plurality of different CPDs.

[0088] Any of the above implementations of the method may include computing (e.g. at **610**, FIG. **6**) a constellation scaling factor for the PCE based on the peak power constraint and the peak enhancement property. In some implementations, the constellation scaling factor may be a distance between adjacent levels of a pulse amplitude modulation (PAM) format.

[0089] In any of the above implementations of the method, the digital processor may be a part of the optical transmitter. In some other implementation, generating of the target CPD may be performed using the digital processor of a computer system separate from the optical transmitter.

[0090] The examples described above, e.g. in the summary section and with reference to any one or more of the FIGS. **1-9**, further provide a method that comprises generating, with a digital processor (e.g. **140**, FIG. **1**; **200**, FIG. **2**), a target CPD (e.g. **430**) for a PCE (e.g. **142**, **210**) of an optical transmitter (e.g. **110**), wherein the generating comprises executing a system performance optimization algorithm (e.g. **600**, FIG. **6**) to search for a test CPD satisfying a system performance criterion (e.g. at **630**). Executing the system performance optimization algorithm comprises:

determining, for the test CPD (e.g. **601**), a constellation scaling factor (e.g. **301**) based on a peak power constraint (e.g. **605**) of a drive signal (e.g. **151**) of the optical transmitter, and computing a system performance estimate (e.g. equation (7)) for the test CPD using the constellation scaling factor, the system performance estimate being computed for a system (e.g. **100**) comprising the optical transmitter connected by a passive optical link (e.g. **130**) to an optical receiver (e.g. **120**).

[0091] The examples described above, e.g. in the summary section and with reference to any one or more of the FIGS. **1-9**, further provide an apparatus comprising a digital processor (e.g. **140**, FIG. **1**; **200**, FIG. **2**). The digital processor is configured to generate a target CPD (e.g. **222**, FIG. **2**; **430**, FIGS. **4, 6, 9**) for a PCE (e.g. **142**, FIG. **1**; **210**, FIG. **2**) of an optical transmitter (e.g. **110**, FIG. **1**). The digital processor is further configured to execute a system performance optimization algorithm (e.g. **600**, FIG. **6**) to search for a CPD satisfying a system performance criterion. The system performance optimization algorithm comprises: determining, for a test CPD, a constellation scaling factor based on a peak power constraint (e.g. **605**) of a drive signal (e.g. **151**, FIG. **1**) of the optical transmitter; and computing a system performance estimate for the test CPD using the constellation

scaling factor, the system performance estimate being computed for a system (e.g. **100**, FIG. **1**) comprising the optical transmitter connected by a passive optical link (e.g. **130**) to an optical receiver (e.g. **120**).

[0092] In some implementations, the digital processor may be further configured to compute a peak value (e.g. **525**) of the drive signal (e.g. **151**) based on the test CPD (e.g. **601**).

[0093] In any of the above implementations of the apparatus, the digital processor may be configured to compute the system performance estimate based on at least one of: signal noise of the optical receiver, and signal noise of the optical transmitter.

[0094] In any of the above implementations of the apparatus, the digital processor may be a part of the optical transmitter. In other implementations, the apparatus may comprise a computing system (e.g. **800**) including the digital processor and configured to communicate the target CPD to the optical transmitter.

[0095] The functions of the various elements shown in the figures, including any functional blocks labeled or referred to as “processors” and/or “controllers,” may be provided through the use of dedicated hardware as well as hardware capable of executing software in association with appropriate software. When provided by a processor, the functions may be provided by a single dedicated processor, by a single shared processor, or by a plurality of individual processors, some of which may be shared. Moreover, explicit use of the term “processor” or “controller” should not be construed to refer exclusively to hardware capable of executing software, and may implicitly include, without limitation, digital signal processor (DSP) hardware, network processor, application specific integrated circuit (ASIC), field programmable gate array (FPGA), read only memory (ROM) for storing software, random access memory (RAM), and non-volatile storage. Other hardware, conventional and/or custom, may also be included. Similarly, any switches shown in the figures are conceptual only. Their function may be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic, or even manually, the particular technique being selectable by the implementer as more specifically understood from the context.

[0096] As used in this application, the term “circuitry” may refer to one or more or all of the following: (a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry); (b) combinations of hardware circuits and software, such as (as applicable): (i) a combination of analog and/or digital hardware circuit(s) with software/firmware and (ii) any portions of hardware processor(s) with software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions); and (c) hardware circuit(s) and or processor(s), such as a microprocessor(s) or a portion of a microprocessor(s), that requires software (e.g., firmware) for operation, but the software may not be present when it is not needed for operation.” This definition of circuitry applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term circuitry also covers an implementation of merely a hardware circuit or processor (or multiple processors) or portion of a hardware circuit or processor and its (or their) accompanying software and/or firmware. The term circuitry also covers, for example and if applicable to the particular claim element, a baseband integrated circuit or processor integrated circuit for a mobile device or a similar integrated circuit in server, a cellular network device, or other computing or network device.

[0097] It should be appreciated by those of ordinary skill in the art that any block diagrams herein represent conceptual views of illustrative circuitry embodying the principles of the disclosure.

[0098] Furthermore in the description above, for purposes of explanation and not limitation, specific details are set forth such as particular architectures, interfaces, techniques, etc. in order to provide a thorough understanding of the present inventions. In some instances, detailed descriptions of well-known devices, circuits, and methods are omitted so as not to obscure the description of the present invention with unnecessary detail. Thus, for example, it will be

appreciated by those skilled in the art that block diagrams herein can represent conceptual views of illustrative circuitry embodying the principles of the technology. All statements herein reciting principles, aspects, and embodiments of the invention, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future.

[0099] While the present invention has been particularly shown and described with reference to example embodiments as illustrated in the drawing, it will be understood by one skilled in the art that various changes in detail may be affected therein without departing from the spirit and scope of the invention as defined by the claims.

Claims

1. A method comprising: generating, with a digital processor, a target constellation probability distribution (CPD) for a probabilistic constellation encoder (PCE) of an optical transmitter; and configuring the PCE of the optical transmitter based on the target CPD; wherein the generating comprises computing the target CPD based in part on a peak power constraint on a drive signal of the optical transmitter.
2. The method of claim 1 wherein the computing comprises selecting the target CPD from a plurality of different CPDs based on a peak enhancement property of a drive circuit of the optical transmitter and the peak power constraint.
3. The method of claim 2 wherein the selecting is based on estimates of transmission performance of an optical system comprising a passive optical link connecting the optical transmitter to an optical receiver for corresponding ones of the plurality of different CPDs, and wherein the estimates are computed based on the peak enhancement property of the drive circuit of the optical transmitter and the peak power constraint.
4. The method of claim 2 comprising mapping the plurality of different CPDs to values of the signal peak enhancement property.
5. The method of claim 4 wherein the mapping includes configuring the digital processor to generate an input signal for the drive circuit for a given CPD, and to output a peak value of a corresponding output signal of the drive circuit.
6. The method of claim 5 including using a set of training CPDs to configure the digital processor for the mapping.
7. The method of claim 5 comprising providing the input signal to a computer model of the drive circuit of the optical transmitter.
8. The method of claim 5 comprising providing the input signal to the drive circuit of the optical transmitter.
9. The method of claim 3 comprising computing the estimates of transmission performance based in part on receiver noise of the optical receiver.
10. The method of claim 3 comprising executing a conditional search algorithm to generate at least some of the plurality of different CPDs.
11. The method of claim 3 wherein the selecting comprises computing a constellation scaling factor for the PCE based on the peak power constraint and the peak enhancement property.
12. The method of claim 11 wherein the constellation scaling factor is a distance between adjacent levels of a pulse amplitude modulation (PAM) format.
13. The method of claim 1 wherein the digital processor is a part of the optical transmitter.
14. The method of claim 1 wherein the generating is performed using the digital processor of a computer system separate from the optical transmitter.
15. A method comprising: generating, with a digital processor, a target constellation probability distribution (CPD) for a probabilistic constellation encoder (PCE) of an optical transmitter;

wherein the generating comprises executing a system performance optimization algorithm to search for a CPD satisfying a system performance criterion, and wherein the executing comprises: determining, for a test CPD, a constellation scaling factor based on a peak power constraint of a drive signal of the optical transmitter; and computing a system performance estimate for the test CPD using the constellation scaling factor, the system performance estimate being computed for a system comprising the optical transmitter connected by a passive optical link to an optical receiver.

16. An apparatus comprising: a digital processor configured to generate a target constellation probability distribution (CPD) for a probabilistic constellation encoder (PCE) of an optical transmitter; wherein the digital processor is further configured to execute a system performance optimization algorithm to search for a CPD satisfying a system performance criterion, the system performance optimization algorithm comprising: determining, for a test CPD, a constellation scaling factor based on a peak power constraint of a drive signal of the optical transmitter; and computing a system performance estimate for the test CPD using the constellation scaling factor, the system performance estimate being computed for a system comprising the optical transmitter connected by a passive optical link to an optical receiver.

17. The apparatus of claim 16 wherein the digital processor is further configured to compute a peak value of the drive signal based on the test CPD.

18. The apparatus of claim 16 wherein the digital processor is configured to compute the system performance estimate based on at least one of: signal noise of the optical receiver and signal noise of the optical transmitter.

19. The apparatus of claim 16 wherein the digital processor is a part of the optical transmitter.

20. The apparatus of claim 16 comprising a computing system including the digital processor and configured to communicate the target CPD to the optical transmitter.
